A070 – Hilbert Space Bridge: Bijective Translation FFGFT to QM

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Contents

.1 Purpose

[STIPULATION] This document formulates the *complete translatability* between the FFGFT mode formalism and standard Hilbert-space quantum mechanics. The translation is not an approximation and not a specialisation, but a **bijective structural statement**: every QM computation is mappable into FFGFT language, and every FFGFT statement about quantum systems is reproducible in Hilbert-space language.

Layer note. Alongside the algebraic bridges [B] and geometric core statements [K], this document contains a substantial proportion of conceptual and ontological framings [S]. The [S] statements concern interpretational questions (Born rule, determinism, Bloch-sphere position) – they are correctly marked, but the reader should know: the ratio here is K : S $\approx$ 1 : 5. The established core is narrow; the rest is framing. That is not a deficiency – framings belong in a bridge document. But bridge and core are not the same.

This is the methodological point: FFGFT does not *replace* quantum mechanics but *extends* it. Both formalisms deliver the same measurable results – up to a ξ-correction of order 10^{-5}, which currently lies below the NISQ noise floor (A160). They differ only in the ontological reading of some basic concepts. One who computes with Hilbert-space tools can continue to do so; the translation table below is the bridge.

The canonical Hilbert space of the full theory is

$$\boxed{\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3}$$

– square-integrable functions on the 4-torus, tensored with the three-element Z_3 fibre space (A030). This document builds it in two directions: *downward* (FFGFT → standard Hilbert space, with reductions) and *upward* (which extensions the standard Hilbert space needs to natively carry the full FFGFT).

.2 Where We Are Geometrically

[B] Before the translation tables, the geometric picture. The full FFGFT geometry is the 4-torus $T^4 = T^3 \times S^1$ (A030): three spatial winding directions and one temporal. For a single qubit – a two-level system at fixed time, without mass dynamics – most of these directions are inactive. Two rotational degrees of freedom remain: one for the position between $|0\rangle$ and $|1\rangle$ (this becomes z), one for the phase (this becomes θ). Their natural space is the 2-torus $T^2 = S^1 \times S^1$.

Pulling one major radius large, T^2 becomes locally the cylinder $\mathbb{R} \times S^1$: one loop is cut, z becomes a *linear* axis. Then contracting the two poles to points, the cylinder becomes the Bloch sphere S^2 of standard QM. This creates a hierarchy of reductions:

Geometry	Dim.	Use
4-torus T^4	4	full FFGFT (masses, α)
3-torus T^3	3	mode space at fixed time
2-torus T^2	2	single qubit, without reduction
Cylinder $\mathbb{R} \times S^1$	2	limit $R \to \infty$, local approximation
Bloch sphere S^2	2	poles contracted, standard QM

[B] What the reductions cost – and where the loss really sits. Each level loses something: $T^4 \to T^3$ the time winding and hence the masses; $T^3 \to T^2$ a spatial winding and hence the third generation; $T^2 \to$ cylinder *one compact loop* – the weakest reduction, and precisely here sits the fractal correction factor $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ (A040): K_{frak} is the cylinder's memory of its torus origin. The loss is *not* the subsequent SI conversion: the duality-carrying decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T} \cdot m = 1$, A060/A090) is a mode-preserving embedding and lossless; compactness survives as a discrete spectrum and as a physical observable (revival time, CHSH$\sim \xi$). The lossy operations are the *reverse direction* $\mathbb{R} \to S^1$, the spatial projections, and – for a readout – the finite, non-periodic measurement window.

.3 States: the Bijective Bridge

[B] In standard QM a qubit is a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \qquad \alpha, \beta \in \mathbb{C}.$$

In FFGFT the same qubit is a point (z, r, θ) on the cylinder/torus:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \qquad z^2 + r^2 = 1,$$

with z as projection onto the computation axis, r as radial superposition amplitude, θ as phase. Both representations are linked by a bridge:

$$\boxed{\alpha = \sqrt{\tfrac{1+z}{2}}\, e^{i\theta/2}, \qquad \beta = \sqrt{\tfrac{1-z}{2}}\, e^{-i\theta/2}}$$

and conversely $z = |\alpha|^2 - |\beta|^2$, $r = 2|\alpha\beta|$, $\theta = \arg(\alpha) - \arg(\beta)$. Consistency is immediate: $|\alpha|^2 + |\beta|^2 = 1$ and $z^2 + r^2 = 1$; $|0\rangle$ corresponds to $(1, 0, *)$, $|1\rangle$ to $(-1, 0, *)$.

[S] The conceptual difference: in QM $|\alpha|^2$ is a *probability* (Born rule), in FFGFT $r^2 = 1 - z^2$ is an *energy density* of the radial configuration. Numerically both coincide at large statistics; FFGFT is a deterministic geometric model whose statistical results agree with the Born rule.

.4 Operators: Pauli Matrices as Rotations

[B] The three Pauli matrices generate all single-qubit operations, with algebra $\sigma_i\sigma_j = \delta_{ij}\mathbb{1}+i\epsilon_{ijk}\sigma_k$. On the Bloch cylinder they correspond to three geometric rotations: σ_z the phase rotation $\theta \to \theta + \pi$, σ_x a (z, r)-rotation by π (bit flip), σ_y the orthogonal rotation with phase flip.

Operator	**Hilbert-space matrix**	**FFGFT geometry**
σ_z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi$
σ_x	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r)-rotation by πK_{frak}
σ_y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Rotation with phase flip
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	swap $z \leftrightarrow r$, $\theta \to \theta + \pi/2$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi/4$

[K] The *only* FFGFT-specific deviation is the factor $K_{\text{frak}} \approx 0.9867$ in the σ_x rotation. It follows from the fractal dimension $D_f = 3 - \xi$ (A040) and is a testable prediction: all π-gates deviate systematically by $\Delta \approx 0.013\,\%$ from the ideal value. In the Hilbert-space formalism this would be an ad hoc correction to be built in as hardware noise; in FFGFT it follows from the geometry.

.5 Gates and Universality

[B] Every unitary single-qubit gate $U \in U(2)$ is a Bloch rotation,

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha}\big(\cos(\phi/2)\,\mathbb{1} - i\sin(\phi/2)\,\hat{n}\cdot\vec{\sigma}\big),$$

in FFGFT the rotation of the (z, r, θ) point around axis $\hat{n}$ by ϕ. The CNOT gate translates as geometric conditioning: *if $z_A = -1$ (control qubit on $|1\rangle$), apply bit flip on qubit B* – topologically a linking of two cylinders (entanglement). Since the universal set $\{H, T, \text{CNOT}\}$ (Solovay–Kitaev) is fully geometrically representable, FFGFT is *universal for quantum computation*: there is no QM statement not formulable in FFGFT.

.6 Tensor Products, Bell States, CHSH

[B] For n qubits the Hilbert space is $\mathscr{H}_n = \bigotimes_{i=1}^{n} \mathbb{C}^2$ with dimension 2^n. In FFGFT n qubits correspond to n mode configurations (z_i, r_i, θ_i) on a shared torus geometry; entanglement is a *topological connection* – knots not reducible to product form. The Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a topologically connected cylinder configuration. Both formalisms reproduce the exponential growth $\dim = 2^n$ with $2 \cdot 2^n - 2$ real parameters.

[K] CHSH prediction. Standard QM gives the Tsirelson bound $|\langle \text{CHSH} \rangle| \leq 2\sqrt{2}$. FFGFT predicts a small, geometrically derived deviation $\Delta\text{CHSH} \approx 10^{-5}$ – the *elementary* deviation per measurement. It lies below the current hardware noise floor and is an open task for high-precision Bell tests (A160/A210). Distinct from this is the cumulative value over many measurements; the two quantities are not directly comparable.

.7 Measurement as Pole Transition

[S] In standard QM a measurement in the computation basis gives value 0 with probability $|\alpha|^2$, with collapse to $|0\rangle$ or $|1\rangle$. In FFGFT measurement is *not* a probabilistic collapse but a geometric interaction driving the (z, r, θ) point to the nearest stable extreme ($z = +1$ or $z = -1$) – a deterministic motion in mode space that is statistically indistinguishable from Born-rule behaviour because of the unknown phase θ. At large statistics:

$$P_{\text{FFGFT}}(z = +1) = \tfrac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0).$$

The statistical results are identical; the Bell-test debate over local realism dissolves geometrically, because entanglement is represented as topological connection, not as non-local correlation.

.8 Unitary Evolution: Schrödinger as Low-Energy Limit

[B] The QM time evolution $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$ is in FFGFT the deterministic motion of the mode configuration $\varphi_{n,l,j}$ on $T^3 \times \mathbb{R}$. The FFGFT mode equation reduces in the low-energy limit ($E \ll E_\xi$) to the Schrödinger equation:

$$i\hbar \frac{\partial \varphi_{n,l,j}}{\partial t} \xrightarrow{E \ll E_\xi} -\frac{\hbar^2}{2m}\nabla^2 \varphi_{n,l,j} + V\varphi_{n,l,j}.$$

At higher energies ξ-corrections appear, which in the Hilbert-space formalism appear as Hamiltonian modifications.

.9 What Is Translatable and What Remains FFGFT-Specific

[B] *All* Hilbert-space statements are mappable into FFGFT: states, operators, gates (with or without K_{frak}), tensor products, entanglement, Bell/GHZ states, measurement, unitary evolution, algorithms (Shor, Grover, VQE). And conversely: the (z, r, θ) configuration corresponds uniquely to the (α, β) vector, geometric rotations to unitary operators, topological entanglements to entangled Hilbert-space vectors. Every computation in one formalism is executable in the other, with identical result.

[K] The translatability holds for *statements about quantum systems*, not for the *input values*. In Hilbert space ξ, the fermion masses, α, and K_{frak} appear as external inputs; in FFGFT they follow from the geometry (ξ as stipulation A020; the masses from $m_i = r_i\,\xi^{p_i} v$, A100/A110; α from A130; K_{frak} from A040). This is the structural asymmetry: Hilbert space has all the tools for QM computations, but no explanation for the input values; FFGFT delivers both from one geometry. In this sense QM is a subset of FFGFT – restricted to the question of *how* systems behave, without the question of *why* the constants have these values.

.10 Operational Equivalence and Interpretative Divergence

[S] Complete translatability must not be read as ontological identity. *Operationally* FFGFT and standard QM agree in every measurable result, up to ξ-corrections. *Interpretatively* they diverge at points where the standard reading makes statements that in FFGFT appear as artefacts of the chosen carrier assumptions. Three points within QM:

- **Born rule as irreducible randomness.** Standard reading: $|\psi|^2$ as ontological probability, the outcome fundamentally random. FFGFT reading: same distribution, but from deterministic pole-transition dynamics on T^4; stochasticity arises at the measurement apparatus, not in the natural process. Operationally identical.
- **Non-locality as ontological property.** Standard reading of Bell correlations: action at a distance in $\mathbb{R}^3$. FFGFT reading: entanglement is a topological connection on T^4 – geometric proximity in a closed geometry, not action at a distance in open space (A160). Results identical, ontological reading structurally different.
- **$\hbar$ as ontological primitive.** Standard reading: $\hbar$ is a fundamental constant of nature. FFGFT reading: $\hbar$ is an SI conversion factor between energy and time measures, following from the closure condition on T^4 (A080/A085). Numerically identical, ontologically derived.

.11 The Four Extensions: Hilbert Space Upward

[B] The downward translation loses a structural layer at each reduction. The reverse question is: which extensions must one *add* to the standard Hilbert space so that it natively carries the full FFGFT? Remarkably, each extension corresponds to an established mathematical structure – FFGFT invents no new mathematics, but is a specific combination of existing structures.

Extension 1 – deformed algebra $SU_q(2)$. The isotropic $SU(2)$ algebra $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$ becomes anisotropic in FFGFT: $\sigma_x^{\text{FFGFT}} = K_{\text{frak}}\,\sigma_x$, while σ_y, σ_z are undisturbed. This is the q-deformation (Drinfeld–Jimbo 1985) with

$$\boxed{q = e^{i\pi(1-K_{\text{frak}})} = e^{i\pi\cdot 100\xi} = e^{i\pi/75} \approx 0.999 + 0.042\,i}$$

The deformation phase is small ($\pi/75 \approx 0.042$ rad) but structurally significant: it breaks spherical symmetry to an axis-distinguished algebra – the cylinder's memory of the torus. Cost: the Hilbert space remains $\mathbb{C}^2$, the 2×2 matrices remain; only the structure constants change. Consequence: all π-gates deviate by $0.013\,\%$ (testable).

Extension 2 – cyclic winding quantum number. The cylinder has one compact loop; the 2-torus has two. To represent the second, the Hilbert space carries a quantum number $n \in \mathbb{Z}$: from $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ one gets $|\psi; n\rangle = \alpha|0; n\rangle + \beta|1; n\rangle$, and

$$\boxed{\mathscr{H}^{T^2}_{\text{FFGFT}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}^2_n}$$

a countably infinite-dimensional space. The winding excitations have energy $E_n \sim n^2\hbar^2/(mR^2)$; at low energies only $n = 0$ is occupied and the space effectively reduces to $\mathbb{C}^2$ – the standard Bloch sphere.

Extension 3 – bundle structure instead of tensor product. In standard QM a spin-$\frac{1}{2}$ particle is $\mathscr{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ with $[L_i, S_j] = 0$; spin-orbit coupling $H_{\mathrm{SO}} = \xi(r)\,\vec{L}\cdot\vec{S}$ must be introduced as an additional term with a free parameter. In FFGFT both position *and* spin arise from windings on the same T^3; the correct Hilbert space is *not* a tensor product but the section space of a spinor bundle,

$$\boxed{\mathscr{H}^{T^3}_{\mathrm{FFGFT}} = L^2(T^3, S)}$$

with $[L_i, S_j] \neq 0$ (corrections $\sim \xi$). Spin-orbit coupling is thereby a *geometric* property of the bundle, not a free parameter.

Extension 4 – temporal winding k_t. In standard QM time is a continuous linear axis with $E = \hbar\omega$. In FFGFT it is a compact winding direction with $k_t \in \mathbb{Z}$ and

$$\boxed{E = \hbar f(k_t)}$$

for a periodic function f; in the lowest sector $E \approx \hbar\omega$. The central statement: the *mass* of a mode is its k_t-winding energy, $m_i c^2 = \hbar f(k_t^{(i)})$, which together with the Yukawa form $m_i = r_i\,\xi^{p_i} v$ (A100) delivers all fermion masses from the winding tuples.

.12 The Structural Point: Known Mathematics, New Combination

[B] Each of the four extensions corresponds to a well-known structure:

Ext.	Mathematics	Established since
1	Quantum groups $SU_q(2)$	Drinfeld/Jimbo 1985
2	Winding excitations	String theory, Kac–Moody
3	Spinor bundles	Differential geometry 1950s
4	Compact extra dimension	Kaluza 1921, Klein 1926

None of these structures is new. What makes FFGFT specific is the **combination**: all four together, with a single geometric justification (T^4 and ξ).

.13 The Full Hilbert Space

[B] With all four extensions one obtains a complete FFGFT-equivalent formulation. In the canonical A-series notation the base space is

$$\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3,$$

where the $\mathbb{C}^3$ factor carries the three-element Z_3 identification (A030); the full formulation supplements it with the spinor bundle S and the deformed $SU_q(2)$ algebra. In this language the FFGFT results are directly derivable from the Hilbert-space structure: the masses from the k_t-windings (A100), α from an eigenvalue relation (A130), spin-orbit coupling from the bundle structure, the K_{frak} corrections from the quantum-group deformation (A040).

[K] On the same space the mass operator acts as a Z_3 circulant; from its diagonalisation follow the Koide condition $Q = 2/3$ and the angle $\theta = 2/9$. This computation is *not* carried out here but belongs substantively in the lepton sector: the operator and its diagonalisation are in A110, the angle $\theta = 2/9$ in A120. This document only provides the space on which they act.

.14 Quantum Graphity as Sketched Subset

[S] The same subset strategy – Bloch sphere ⊂ FFGFT through reduction – can be applied to another established, published quantum gravity theory: *Quantum Graphity* (Konopka, Markopoulou, Severini, Smolin; *Phys. Rev. D* 77, 104029, 2008; arXiv:0801.0861) – a background-independent model in which space and geometry emerge from a dynamical weighted graph. It is mathematically clearly defined and a concrete comparison object. It is a comparison with established physics, not a load-bearing argument of FFGFT; and – the crucial difference from the Hilbert-space bijection above – the reduction is *not carried out* but a hypothesis. Hence the consistent $[S]$ marking throughout.

Five reduction steps: (1) continuum to discrete (triangulation of T^4, not proved); (2) topology forgotten (K_N vs. T^4 winding structure); (3) windings to spin labels (physical meaning of (n_x, n_y, n_z, k_t) lost); (4) geometry parameter to phase parameter ($\xi = 4/30000$ fixed vs. free couplings $g_V, g_B, \ldots$); (5) matter sector discarded (all 12 fermion masses absent).

What survives: one genuine correspondence – emergent $U(1)$ from microstructure (Levin–Wen, extended version only). FFGFT contains Hilbert-space QM provably as a subset, and discrete-geometry programmes plausibly as a special case.

.15 Field Theory: Modes, Lagrangian, Schrödinger Limit

[B] The complete field theory of the Hilbert space – explicit mode functions $\phi_{n,l,j}$, free Lagrangian $\mathscr{L}_0$, bridge formula Lagrangian ↔ mode operator, Dirac line, and Schrödinger limit – is in A075. Here only the key formulas needed for the Hilbert-space connection.

[B] Bridge formula. Lagrangian and mode operator describe the same physics (derivation in six steps: A075):

$$\int \tfrac{1}{2}(\partial\,\delta m)^2\, d^4x = \tfrac{1}{2}\langle\psi|\Phi^2|\psi\rangle, \qquad \Phi = \sum_i m_i P_i.$$

Mass arises not as a Lagrangian parameter but as an eigenvalue of the mode operator.

.16 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$q = e^{i\pi/75}$	Deformation parameter of the Drinfeld–Jimbo quantum group; follows from $q = e^{i\pi\cdot 100\xi}$ with $\xi = 4/30000$
$U_q(\mathfrak{su}(2))$	q-deformed $\mathfrak{su}(2)$ algebra
$[n]_q$	q-number: $[n]_q = (q^n - q^{-n})/(q - q^{-1})$
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices (spin operators)
S_{CHSH}	CHSH value, theoretical maximum $2\sqrt{2} \approx 2.828$

.17 Verification Scripts

- Bijective bridge $(z, r, \theta) \leftrightarrow (\alpha, \beta)$: numerical verification of $|\alpha|^2 + |\beta|^2 = 1$ and $z^2 + r^2 = 1$ over random states, plus operator translation Pauli ↔ rotation: `python/A_Serie_Skripte/a070_tensorfaktor.py`

- q-deformation and factor structure: $q = e^{i\pi/75}$, limit $q \to 1$, and the tensor factor $L^2(T^4) \otimes \mathbb{C}^3$: `python/A_Serie_Skripte/a070_zirkulant_dft.py`

.18 Open Questions

The ΔCHSH $\sim 10^{-5}$ deviation is calculated, not measured. The CHSH scaling formula is fully derived in A165; the computational framework (ϕ-QFT, Bell damping, T0-Shor) and the hardware status are there. The IBM-Heron-r2 measurement (May 2026) confirms Bell violation ($S = 2.74$, 96.9 % of the Tsirelson bound), but the device-to-device variation ($\sim 1.4\,\%$) exceeds the ξ effect ($\sim 0.001\,\%$) by a factor ~ 1400 – not resolvable with today's NISQ hardware in principle (A165). As a prediction with falsification criteria, ΔCHSH stands in A210.

Top quark, Down-type quarks, and quark r_i deep structure are unclear. The lepton exponents and Up-type quarks (up to Charm) follow the geometric $2/3$ sequence (A075/A100); the top quark ($p_t = -1/3$, no known geometric reason) and the Down-type arithmetic step size remain open (A150).

.19 Supersedes

This document subsumes: Dok. 230 (translatability FFGFT ↔ Hilbert-space QM, bijection, operators, gates, measurement, interpretative divergence), Dok. 231 (the four Hilbert-space extensions), Dok. 232 (Quantum Graphity as sketched subset), and the framework section of Dok. 282 (the Hilbert space explicit). Incorporated: R41 (location of losses: the type-I decompactification is lossless; lossy are the reverse direction, spatial projection, and measurement window). *Not* taken over: the mass operator/Koide part of Dok. 282 belongs substantively in A110; the HLV/CP comparison part of Dok. 282 remains independent as an IPI framework comparison (A240).

.20 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.